

ZeToD – Zero To Deploy

Know exactly where you stand. Beginner, Intermediate, or Job Ready.

ZeToD is an open source Python skill assessment platform built for African developers. It measures a developer's accuracy, speed, and knowledge depth against an expert benchmark – giving them a live skill score and a clear picture of where they stand.

The Problem

Thousands of African developers are learning Python but have no honest way to know if they are job ready. Certificates mean nothing. Self-assessment is unreliable. ZeToD fixes that.

Live URLs

Resource	URL
Frontend	https://zetod-platform.vercel.app

Resource	URL
Backend API	https://zetod-backend.onrender.com
Health Check	https://zetod-backend.onrender.com/health

What It Does

- Assesses Python knowledge across Beginner, Intermediate, and Expert levels
 - Combines theoretical questions and practical coding challenges
 - Measures accuracy, speed, and knowledge depth
 - Assigns a clear level – Beginner, Intermediate, or Job Ready
 - Recommends specific topics to study if not yet job ready
 - Provides a live skill score visible on the dashboard
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Tech Stack

Layer	Tool	Hosting
Frontend	React + Tailwind CSS + Vite	Vercel
Backend	Node.js + Express	Render
Database	PostgreSQL	Supabase
Auth	Supabase Auth (Email + Google + GitHub OAuth)	Supabase
Analytics	PostHog	EU Cloud
Error Tracking	Sentry	Cloud
CI/CD	GitHub Actions	GitHub
Code Execution	Judge0 (planned)	Self-hosted

Getting Started (Local Development)

Prerequisites

- Node.js 18+
- npm

Frontend

```
git clone
https://github.com/Kennison7/ZETOD-
Platform.git
cd ZETOD-Platform
npm install
```

Create a `.env` file in the root:

```
VITE_SUPABASE_URL=your_supabase_url
VITE_SUPABASE_ANON_KEY=your_supabase_anon_k
ey
VITE_POSTHOG_HOST=https://eu.posthog.com
VITE_API_URL=https://zetod-
backend.onrender.com
```

```
npm run dev
```

Backend

```
cd backend
npm install
```

Create a `.env` file in `/backend`:

```
PORT=5000
SUPABASE_URL=your_supabase_url
SUPABASE_ANON_KEY=your_supabase_anon_key
SUPABASE_SERVICE_KEY=your_supabase_service_
```

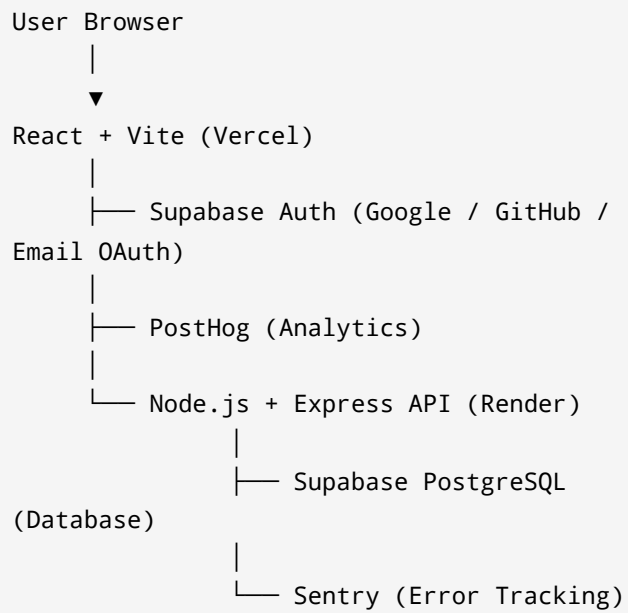
```
key
JWT_SECRET=your_jwt_secret
```

```
npm start
```

Run Tests

```
cd backend
npm test
```

Architecture



CI/CD Pipeline

GitHub Actions runs on every push to `main` :

- Installs dependencies
- Runs 10 Jest/Supertest tests
- All tests must pass before merge

Current test status: 10/10 passing

Build Progress

Week	Focus	Status
Week 1	Problem definition, architecture, repo setup	✓ Done
Week 2	Backend deployment, Jest tests, CI/CD pipeline	✓ Done
Week 3	Supabase auth, JWT middleware, PostHog, Sentry, rate limiting	✓ Done
Week 4	Google + GitHub OAuth, Vercel env vars, Lighthouse scores, 20 user signups	✓ Done

Week	Focus	Status
Week 5	OAuth bug fix, security hardening, OWASP pass, JWT rotation	✅ Done
Week 6	Production-grade build, demo video, final submission	🔄 In Progress

Security

- HTTPS enforced on all endpoints (Vercel + Render)
 - JWT secret stored in GitHub Secrets only
 - No hardcoded credentials in repository
 - Rate limiting on all auth endpoints
 - Input validation on register and login
 - Debug routes removed from production
 - OWASP basic checklist completed (Week 5)
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Team

Name	Role	GitHub
Kenneth Omoregie	Project Lead + Backend	@Kennison7
Joseph (Brayden)	Frontend	@Brayden-Code-4
Emmanuel Gbafore	CI/CD + QA	@EmmanuelGbafore

License

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